Development of a Collaborative Robotic Tool Handover System for Manufacturing Work Cells

Project Proposal — MREN 403

Submitted to:

Dr. Matthew Pan

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Date:

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Cover Letter

Dear Dr. Pan,

On behalf of our MREN 403 project team, we are pleased to submit this proposal for the Collaborative Robotic Tool Handover System in response to the MREN 403 proposal call.

Manufacturing's workforce is aging, and manual assembly work remains largely unautomated because robots still cannot match the dexterity and judgment of an experienced worker. Our project responds to this gap by developing a cobot assistant that retrieves a requested tool from a rack, delivers it to an operator's hand on voice command and returns the tool to the rack once the operator is finished. This allows the operator to stay focused on the skilled part of the task instead of fetching and putting away tools.

The proposal sets out the system’s scope and its three subroutines, the requirements it must meet and how each will be tested, our approach to safety, and a schedule built around the course design reviews. Beyond a working prototype, the project will deliver a documented platform that future researchers can continue to build on.

We welcome the opportunity to discuss this proposal further and can be reached at *koa.schuette@queensu.ca*.

Sincerely,

Ryan Donen, Ryan Henderson, Koa Schuette, and Tanner Warren

MREN 403 Group 16

1.0 Executive Summary

Canada's manufacturing workforce is aging, and experienced workers are retiring faster than they can be replaced. Automation is often proposed as the answer, but robots still cannot match the dexterity and judgement that skilled assembly work requires, so most of this work remains manual. Rather than replacing skilled workers, this project explores how a collaborative robot can support them by taking over routine tasks around them.

The project will design and build a Collaborative Robotic Tool Handover System using a UR5e robotic arm provided by the Ingenuity Labs Research Institute. Working beside a single operator at a fixed workbench, the system will retrieve a requested tool from a rack on voice command, deliver it directly to the operator's hand, and return it to the rack once the operator places it in a return zone. To do this, the system combines voice command recognition, camera-based tool detection, hand and body tracking, and an electromagnetic end effector that releases the tool when it senses the operator's pull.

The system will be evaluated over 48 delivery and 48 return trials using three screwdrivers and four operators. Its requirements include completing each delivery and return within 15 seconds, achieving a successful handoff in at least 80% of trials, and picking up the correct tool in at least 96% of attempts. Safety measures follow CSA robot safety standards and rely on the UR5e's certified safety functions, with camera-based tracking used only as a supplementary layer.

The project runs from October 2026 to April 2027 and is organized around the course design reviews, with one subroutine completed before each review. The estimated cost is $500, since the robot, cameras, and computing are provided by Ingenuity Labs. Final deliverables include the working system, the trial results, and documentation that allows Dr. Pan and Ingenuity Labs to continue using the platform for handover research after the project ends.

2.0 Problem Statement

2.1 Background

Canada's workforce is aging, and manufacturing is feeling it most. The share of manufacturing workers aged 55 and older rose from 9.8% in 2001 to 24.2% in 2022, the largest increase of any industry [1]. As these workers retire, they take years of hands-on expertise with them, and there are not enough new workers to replace them. In the United States, up to 1.9 million manufacturing jobs could go unfilled by 2033 [2], and only 14% of Gen Z respondents say they would consider industrial work as a career [3].

A common proposal is to fill this gap with robots, and the International Federation of Robotics identifies automation as a key strategy for addressing labor shortages [4]. However, robots have not been able to replace the skill and judgement of experienced workers. Assembly remains predominantly manual, and even in the automotive industry, where robots are well established, only an estimated 20% of assembly is automated [5]. Tasks that require dexterity, flexibility, and adapting to variation are still done by people.

Rather than replacing the worker, a collaborative robot (cobot) can act as their assistant. In this arrangement, the human provides the dexterity and decision-making that robots lack, while the robot provides precision, repeatability, and consistency [5]. However, most cobot applications today stay at the lower levels of collaboration, with the robot and worker completing tasks independently rather than the robot responding to the worker in real time [6]. Tool delivery is one way a cobot can respond to a worker directly. Khawaja et al. developed an assistant robot that follows a worker around an assembly work cell and supplies the parts and tools needed for each task. When the robot predicted where the worker would move next, assembly cycle time dropped by 25% to 34% across four participants [7]. These gains came from the robot working in parallel with the worker, not from the robot being faster than the worker. A robot that brings the requested tool and puts it away afterward applies the same idea to tool handling, so the operator keeps working while the tool is on its way.

2.2 Problem Statement

As Section 2.1 describes, skilled workers are difficult to replace with automation, but a cobot that responds directly can help them work more efficiently. Handing a tool to the operator is a direct form of this support. It is harder than ordinary pick and place, because the robot has to reach a moving hand rather than a fixed point, release the tool only once the operator has a grip on it, and stay within safe speed and force limits while working beside them [8], [9].

The problem this project addresses is that skilled workers lose time and focus retrieving and returning their own tools, and current cobot applications rarely provide this kind of direct, real-time assistance [6].

This project demonstrates a solution on tool handling. It is a collaborative assistant that hands the operator the tool they ask for and returns it to the rack once they are done, so they can stay focused on the work that needs their expertise. Beyond the prototype, the project provides a documented platform for further research into robot to human handover, a capability that broader assistive cobots will depend on.

3.0 Project Scope

This project will design and build a Collaborative Robotic Tool Handover System, a robotic arm that supports a skilled operator by delivering and returning tools on voice command (Section 2.2). The team will build the workbench, tool rack, return zone, and end effector, and will develop the voice, vision, and motion subsystems that run the system on the UR5e. The result will be a working prototype and a documented platform for future handover research.

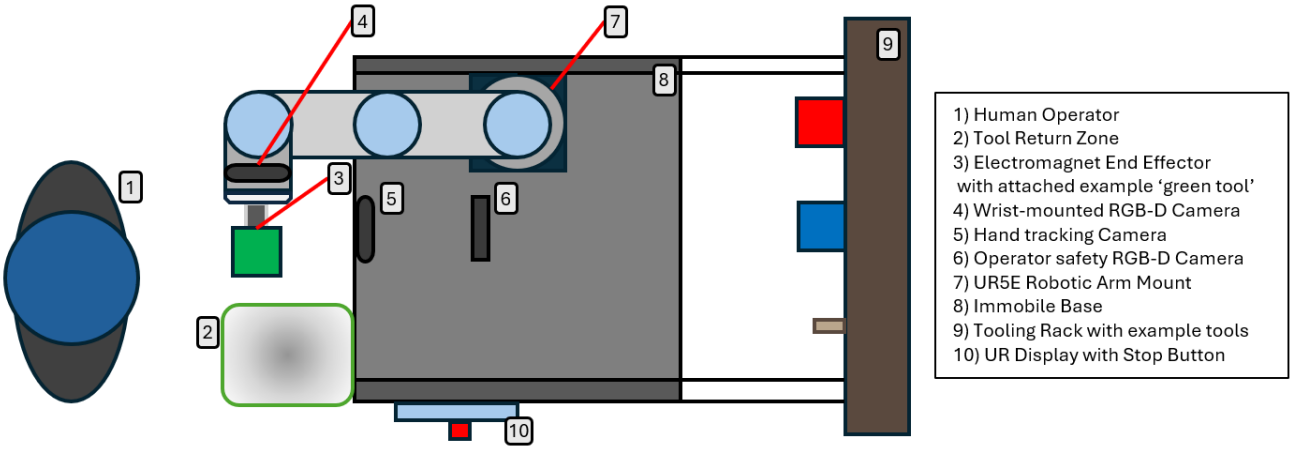


Figure 1: Proposed design of cobot workspace

3.1 Demonstration

The final demonstration will show a single operator working at a fixed bench beside the robot. When the operator needs a tool, they ask for it by voice. The robot retrieves the correct screwdriver from the rack and places it in the operator's hand, without the operator reaching for it or moving to meet the robot. When the operator is finished, they set the tool down in the return zone, and the robot puts it back on the rack. This cycle will be repeated across three different screwdrivers and several operators, following the trial structure in Section 4.

3.2 Subroutines

3.2.1 Rack to Robot

The operator requests a tool by saying a wake word followed by a voice command. The robot identifies the requested screwdriver on the rack and picks it up. If the tool is missing or the pickup fails, it tries again or reports the failure to the operator.

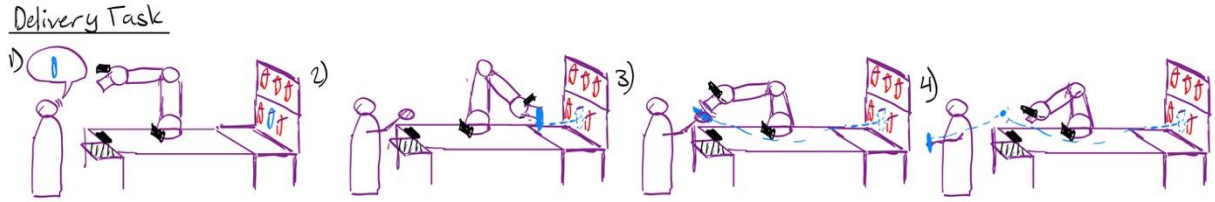


Figure 2: Visual Aid Rack to Robot & Robot to Human

3.2.2 Robot to Human

The robot brings the tool to the operator's hand, slowing as it gets closer to them. Once the tool is presented, the operator grasps it and pulls, and the robot releases it. If the robot loses track of the operator's hand and can't find it again within a set time, it returns the tool to the rack.

3.2.3 Workbench to Robot & Robot to Rack

When the operator is finished with a tool, they place it in the return zone on the workbench. While idle, the robot watches the return zone, and once a tool has rested there for a short time, the robot identifies it, picks it up, and places it back in its assigned position on the rack, ready for the next request.

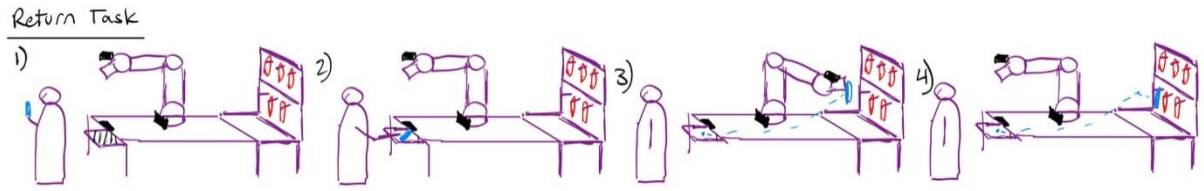


Figure 3: Visual Aid Workbench to Robot & Robot to Rack

4.0 Requirements

4.1 Stakeholders

Three groups have a stake in the project. The operator is the person at the workbench who requests and receives tools. In industry this would be a skilled assembly worker, and in this project the four team members fill the role during all trials. The research users are Dr. Pan and the Ingenuity Labs Research Institute, who are providing the UR5e and cameras and will continue using the platform after the project ends. The course, through the MREN 403 teaching team, sets the deadline, budget, and review milestones. Table 1 lists what each group needs and the requirements those needs produce.

Table 1: Stakeholder needs and the requirements they produce

Stakeholder	Need	Requirement it produces
Operator	Get a tool without stopping work or putting the workpiece down	Voice command; cycle time target
Operator	Not have to move their hand to meet the robot	Hand localization accuracy; definition of a successful handover
Operator	Be safe working beside the robot	Speed and force limits [9]; release only on grip
Dr. Pan and Ingenuity Labs	Continue using the system after the project ends	Documentation and repository (7.0); R7
Dr. Pan	Measured results to compare future methods against	Measured success rate and cycle time from final trials
Course	Finish by April 2027 within budget	Schedule (8.0); budget (9.0)

Each stakeholder's needs in Table 1 were turned into a measurable requirement, listed in Table 2. Every requirement has a minimum that must be met for the system to pass. Each value is justified in one of three ways: a measured human baseline, published results from comparable systems, or the specifications of the hardware being used. Most requirements are checked in a single shared set of trials, described below Table 2, so that one test session covers several requirements at once.

Table 2: List of requirements, their justifications, and the testing plan

Requirements	Justification	Testing Plan
R1: Delivery and return of tool each completed in ≤ 15 s, target 5 s	• Target matches time for a person to do the same task 5s (measured average of 8 people) • Max applies a safety factor of 3x for slower, safety-limited motion	Shared trials; pass if every trial ≤ 15 s

R2: Successful handoff in ≥ 80% of trials, target 90%	• Comparable systems report 73.3% in simulation [10] and 80% on hardware [11] • Known tools, fixed rack, and fixed grasp reduce complexity, justifying a higher minimum	Shared trials; pass if ≥ 39 of 48 successful
R3: Securely hold payloads up to 0.45 kg with no slip during motion	• Screwdrivers weigh ~0.15 kg each • Safety factor of 3 allows heavier tools in future applications	No drops in shared trials, plus one 0.45 kg test object moved through a full handoff motion with no visible slip
R4: Identify, deliver, and return three distinct tools	• Three colored screwdrivers are sufficient to demonstrate tool identification at the prototype stage	Shared trials; each tool to be requested 16 times
R5: Correct tool picked up in ≥ 96% of attempts	• Matches the grasp success of a comparable vision-guided UR5 system [12]	Shared trials; pass if ≤ 2 failures for rack and return zone pickups
R6: Hand Localization	• Leap Motion Controller 2 palm accuracy is ~11 mm across its working volume [13] • Remaining margin covers calibration and robot positioning error	Shared trials; verified through R2 handoff success
R7: Setup and operation fully documented for handover to Ingenuity Labs	• Requested by a key stakeholder, as identified in Table 1 • Allows the system to be used and extended after the project ends	A lab member outside the team runs one full cycle using only the documentation

Trial structure:

The cycle time, handoff, and retrieval requirements are evaluated over 48 trials (4 operators × 3 tools × 4 repetitions) with randomized tool requests, following the structure used in published handover studies [11]. Each delivery trial records cycle time, handoff success, and retrieval success, with a matching set of 48 return trials.

5.0 Industrial Standards and Safety

The project's safety assessment and testing procedures will be informed by CSA Z434:26 for robot applications [14], CSA Z1002:12 for hazard identification and risk assessment practices [15], and CSA Z432:23 for safeguard inspections [16]. The UR5e's built in safety functions are certified to Performance Level d, Cat. 3 under ISO 13849-1 [17] and configurable speed, force, power, momentum, and workspace limits will be selected according to the applications risk assessment.

During testing, a second team member will monitor the experiment and be ready to activate the teach-pendant emergency stop. Emergency-stop behavior will be verified for robot motion and end-effector operation, including whether loss of tool power could release an object or create another hazard. Camera-based operator tracking adds a supplementary layer of safety that will slow or stop the robot when the operator's body or head is in the path of a delivery. This tracking system will not be credited as a safety-rated protective function; the protective measures must remain adequate if tracking fails.

Following the Z1002 process, hazards of the system will be identified, eliminated if possible, and adequate measures will be taken to control remaining risk [15]. The assessment will include collision, trapping, gripping, dropped-object, and perception-failure hazards during setup and testing. Changes affecting safety will be reviewed before testing resumes, and assessment and validation records will be maintained in the testing documentation. Safety functions and safeguards will also be checked at the start of each session by a team member trained on the UR5e safety system.

6.0 Methods

6.1 Workspace and sensor calibration

The tool rack and return zone are rigidly mounted to the robot base as shown in Figure 1, so their positions are known from CAD. To account for differences between modeled and built fixtures, a datum routine is run once after assembly. This routine includes guiding a datum tool on the robot end effector to three corners of the tool rack and return zone. The recorded corners fix each plane's position and rotation in the robot base frame. Once calibrated, both fixtures are treated as static. The wrist RealSense's intrinsic parameters can be found from checkerboard images at ten or more orientations and validated by reprojection error [18]. All three cameras including the eye-in-hand RealSense, the base mounted Ultraleap, and the cell RealSense cameras are related to the base frame from CAD.

6.2 Rack to Robot

The robot listens continuously for a wake word [19] followed by a descriptive request drawn from a set of commands, rejecting low-confidence or unrelated speech inputs [20]. On a valid request it moves to a fixed observation pose in front of the rack, with motions planned and executed through MoveIt [21]. A YOLO model trained on a custom image set of the three screwdrivers detects each tool in the eye-in-hand camera and returns its class, position, metal-end location, and orientation [22]. This can be done with oriented bounding boxes, segmentation, or with keypoints which can yield better grasping information [23]. If the requested tool is detected with sufficient confidence, its position is found using its position within the RGB captured frame and corresponding depth values. The robot approaches the metal end and energizes the electromagnet, and the eye-in-hand camera confirms that the tool is on the magnet before the arm proceeds. Otherwise, the robot attempts again or reports the failure.

6.3 Robot to Human

The Ultraleap sensor on the base platform tracks the operator's hand and supplies the handover target in the robot's base frame [13]. The robot waits until the hand is held steady in the receiving position, then moves the tool toward it. A second RealSense camera monitors the cell and tracks the operator's skeleton. The distance between the operator's body and the robot sets the arm's speed: the robot moves faster while the operator is far away, slows as it approaches, and stops if the operator's body or head enters the delivery path. These speeds stay within the UR5e safety limits set in Section 5.0. An emergency stop remains within the operator's reach. A force sensor in the end effector detects the operator's pull, and a sustained

pull along the handover direction above a threshold de-energizes the electromagnet, releasing the tool. If the hand is lost and not recovered within a timeout, the robot returns the tool to the rack.

6.4 Workbench to Robot

From an idle pose that keeps the return zone in view, the RealSense camera monitors for tools within the calibrated boundaries, utilizing the previously mentioned identification and localization methods. A tool is considered returned when its calculated position remains within the return boundary for longer than a specified time threshold and neither of the operator's hands is detected within that boundary, which prevents triggering while the user is still placing it. The robot aligns the end effector with the tool's axis where it can. Otherwise, it records the grasped orientation from the wrist camera and adjusts the placement orientation on the tool rack accordingly, so the tool is mounted correctly.

7.0 Deliverables

Physical platform. A pegboard tool rack and workbench return zone, both rigidly mounted to the UR5e workstation, three screwdrivers with steel ends for electromagnetic pickup, and mounts for the base Ultraleap and the cell RealSense camera.

End effector. A 12-24 V electromagnet with its driver circuit, the wrist RealSense mount, and force sensing to detect the operator's pull, rated for the 0.45 kg payload.

Software. Voice command recognition, the YOLO tool detection model with its labelled dataset, MoveIt motion planning, hand and skeleton tracking, and the three subroutines in Section 3.2, integrated into the full handover cycle.

Calibration tools. The datum routine for the rack and return zone and the wrist camera calibration procedure described in Section 6.1, so the cell can be recalibrated if a fixture or camera is moved.

Documentation and repository. A setup and operation guide that lets a lab member outside the team run a full cycle without help, and all software, CAD, and data, handed over to Dr. Pan and Ingenuity Labs as one repository.

Reports. The Final Design Report with trial results against each requirement in Section 4 and a final budget statement, along with Engineering Portfolios I and II and the PDR and CDR materials on the schedule in Section 8.0.

8.0 Schedule of Work

Each subroutine is integrated before the design review that presents it, as shown in Table 3. Rack to robot is built first because the other two subroutines need a tool on the magnet, and workbench to robot comes last because it reuses the detection and placement already built for the rack.

Table 3: The Schedule of work for certain milestones along with their target date

Milestone	Description	Target Date
Fabrication complete	Tool rack, return zone, camera mounts, and electromagnet end effector built; electromagnet bench tested at 0.45 kg (R3).	Oct 30, 2026
Engineering Portfolio I	Group portfolio submitted (course deliverable).	Nov 8–14, 2026 (Wk 9)
Rack pickup components working	YOLO tool detection, MoveIt pickup from the rack, and voice commands each tested on their own.	Nov 20, 2026
Rack-to-robot subroutine integrated	Voice request, tool detection, and pickup combined end to end.	Nov 27, 2026
Preliminary Design Review	PDR presentation and report submitted, showing the rack-to-robot subroutine.	Nov 29–Dec 5, 2026 (Wk 12)
Engineering Portfolio II	Group portfolio submitted (course deliverable).	Jan 24–30, 2027 (Wk 4)
Handover components working	Ultraleap hand tracking, cell camera skeleton tracking with speed scaling, and force-triggered release each tested on their own.	Jan 30, 2027
Robot to human integrated	Handover combined with rack to robot, including return to rack if the hand is lost.	Feb 12, 2027
Workbench to robot integrated	Return zone detection, pickup, and placement back on the rack combined and tested.	Feb 26, 2027
Critical Design Review	CDR presentation, updated design, and remaining risks submitted, showing all three subroutines.	Feb 28–Mar 6, 2027 (Wk 8)
Full system integration (feature freeze)	All three subroutines combined into the complete request-retrieve-handover-return cycle; system frozen for testing.	Mar 12, 2027
Trials and Testing	Conducting trials mentioned in Section 4.0 to benchmark system	Mar 12–Mar21, 2027
Demonstration and Final Design Report	Live demonstration and defense; final report, budget statement, and repository submitted.	Mar 21–Apr 3, 2027 (Wk 11–12)

9.0 Budget and Resources

Due to the nature of this project as a technical platform for a research project, several critical pieces of hardware have already been purchased and provided to the team by the Ingenuity Labs Research Institute as shown in Table 4.

Table 4: Itemized project costs and provided equipment

Item	Justification	Estimated Cost
12-24V DC Electromagnet End Effector	The active component in the end effector design. Used to grip metallic tools.	CAD$ 100.00
End Effector mount and magnetic shielding	Materials required for the mounting of the end effector and camera onto the robotic arm, shielding to isolate end effector from camera.	CAD$ 70.00
Force Sensor	Component of end effector system. To be used to determine when the tool is considered 'received' by the human operator during the hand over process.	CAD$ 100.00
Microphone	Provided independently by team. Enables input of vocal commands from human operator for 'Rack to Robot' subroutine.	Owned
Custom Pegboard Tool Rack	Customized design and assembly according to toolset and attachment to fixed base.	CAD$ 200.00
Prototype Fake Tool Set	3D printed tools with steel ends used during early development, before trials with the real tool set.	CAD$ 5.00
Tool Set	Realistic tool set for cobot handovers.	CAD$ 25.00
Universal Robots UR5E Cobot Robotic Arm with Large Robot Workstation	Provided to team by Ingenuity Labs Research Institute. Robotic arm to be used in all stages of the project.	Provided
Wrist-mounted eye-in-hand RGB-D camera	Provided to team by Ingenuity Labs Research Institute. Enables YOLO tool detection through wrist-mounted camera. RealSense D435 Camera or other suitable replacement.	Provided
RealSense D455 RGB-D Camera	Provided to team by Ingenuity Labs Research Institute. Safety-oriented depth camera for tracking human-robot collision potential.	Provided
Ultraleap 'Leap Motion Controller 2' IR Camera with hand-tracking capabilities.	Provided to team by Ingenuity Labs Research Institute. Enables human hand-tracking for tool delivery process.	Provided
Computing Power	Execution of the subroutines will be completed using computing power provided to team by Ingenuity Labs Research Institute. Note, this is not referring to model training.	Provided
Total		**$500.00 total**

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